

Data, Tasks, Marks, and Channels

From Raw Data to Visual Encoding

Information Visualization · Lecture 3

Perception tells us how people see; design decides what to show

- The view's perceptual path tells how people will see the objects in your visualization.
- Now, we need to think how we represent the raw data into visual objects.

Agenda

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- | | | |
|----|--------------------------------|--|
| 01 | Interpret data | Recover meaning from fields and documentation. |
| 02 | Abstract data and tasks | Select items, attributes, relations, actions, and targets. |
| 03 | Encode visually | Map what exists to marks and properties to channels. |
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We start with raw observations

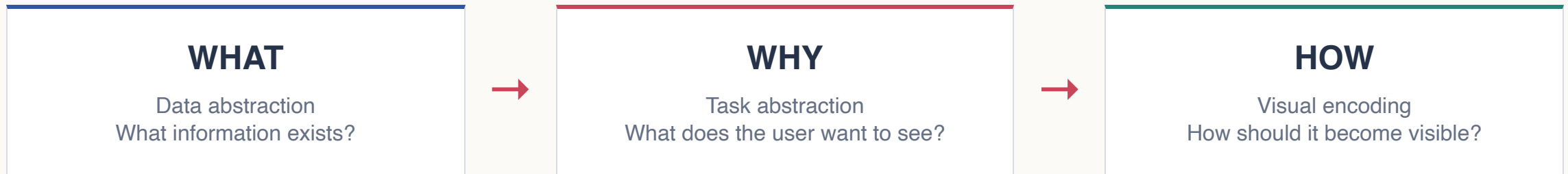
trip_id	started_at	start_station	end_station	duration (min)	rider_type	bike_type	period
50821	2019-06-01 08:30	W 21 St	E 17 St	12	member	classic	before
61204	2020-06-01 08:42	Pershing Sq N	W 21 St	18	casual	classic	during
73519	2021-06-01 08:50	Central Park S	Broadway	9	member	electric	after
50822	2019-06-01 09:05	E 17 St	Central Park S	22	casual	classic	before
61205	2020-06-01 09:12	Broadway	Pershing Sq N	15	member	electric	during

This is what we have. But what should we visualize?

It depends on answers to three questions:

- What information is relevant?
- What does the audience want to see? (Why do they need a visualization for the data?)
- How should it become visible?

Every visualization design moves through What, Why, and How



Raw data does not choose a chart. Interpretation and purpose come first.

PART 01

Interpret data before designing a view

Stored values become useful only after their domain meaning is established.

Data semantics gives stored values meaning

Data semantics is the real-world meaning of data.

Raw data does not directly tell us what to visualize.

start_station	end_station	start_time	duration (min)
1001	1002	08:30	12
1005	1001	08:42	18
1008	1006	08:50	9

For example, what does 1001 represent?

- a value?
- a rank order?
- an identifier?

Data semantics tells us what values represent

- **Column names**, metadata, documentation, and **domain knowledge** provide the context needed to interpret a value.
 - For example, the column name `station_id` tells us that `1001` is an identifier.
- Stored value + context = meaning
- Once the meaning is clear, we can decide what matters for the task.

PART 02

Abstract the data for the task

Identify the entities, properties, relations, and comparisons that matter.

Data abstraction depends on meaning and task

Data abstraction is a task-dependent description of the items, attributes, and relations relevant to a visualization problem.

TASK

How does trip duration vary over time?

start_station	end_station	start_time	duration (min)
1001	1002	08:30	12
1005	1001	08:42	18
1008	1006	08:50	9

1

Data semantics

Column names + Citi Bike documentation tell us:

- each record is a bike trip
- `started_at` is the trip start time
- `duration` is elapsed minutes
- stations are trip endpoints

+

2

Task

How does trip duration vary over time?

The task focuses attention on trips, start time, and duration.

=

3

Data abstraction

Items: trips

Attributes: start time, duration

Relations: none required

- The task selects which entities, properties, and relations matter.
- Change the task, and the useful abstraction may change.

The item determines which attributes and relations matter

An item is an entity we want to reason about in a visualization.

Stored record

start_station	end_station	start_time	duration (min)
1001	1002	08:30	12
1005	1001	08:42	18
1008	1006	08:50	9

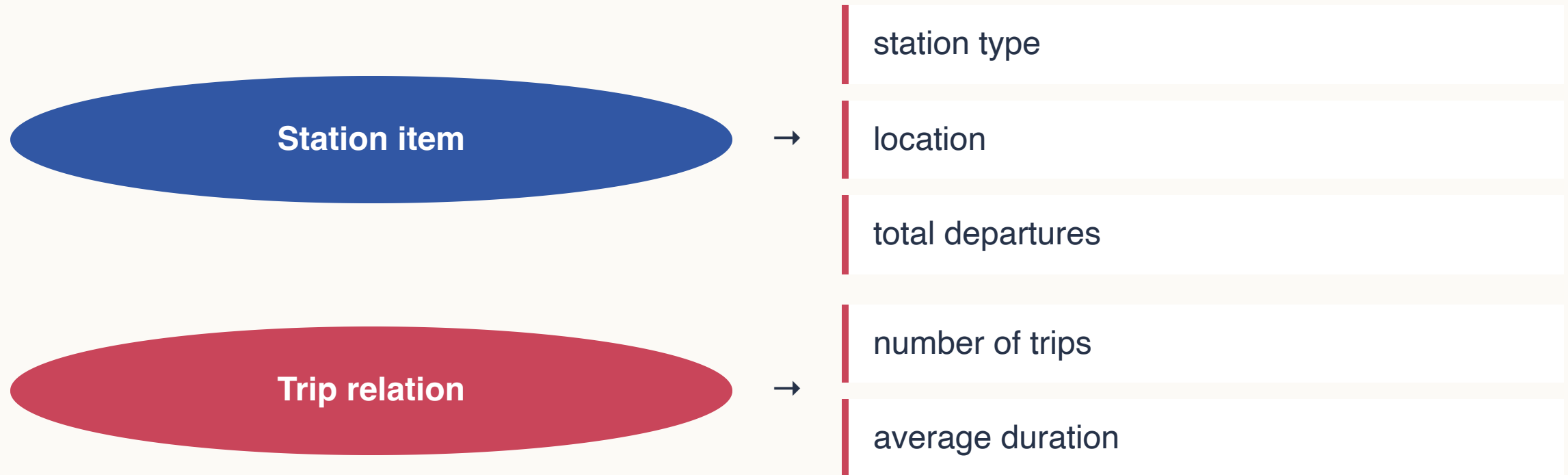
Two tasks, two conceptual data abstractions

Task	Item	Attributes	Relations
Duration over time	TRIP	start time duration	none required
Station connectivity	STATION	station name location total departures	trips connect station pairs Relation attributes: trip count, average duration

- Row $\neq$ item: A row is a data representation; an item is a task-dependent abstraction.
- Transform the data when the two do not align.

Attributes describe items—and relations

Items and relations are the objects represented in a visualization. **Attributes** are properties of those objects. When encoded, attributes determine what the objects look like.



Attribute types describe the comparisons that make sense

We perceive information in a visualization by **comparing values**. We distinguish three types of attributes, and each supports a different kind of comparison.

Categorical

Equality, with no inherent order

country

China $\neq$ Brazil

Ordinal

Equality + order; gaps need not be arithmetic

T-shirt size

S < M < L

Quantitative

Meaningful magnitude; arithmetic comparisons make sense

population

20M is twice 10M

Quick classification

What type is each attribute?

Country	Religion	Population	Student ID
Rank	Temperature	Satisfaction	Year

Pairs · 3 minutes · Categorical / ordinal / quantitative · Defend ambiguous cases

How do we determine an attribute type?

Ask: Given what this attribute means and what the task requires, what comparisons should the visualization preserve?

Decision rule

Only equality / difference matters	→ CATEGORICAL
Order matters, but distance does not	→ ORDINAL
Numerical difference / magnitude matters	→ QUANTITATIVE

Example: Year

Distinguish different cohorts Only identity matters → Categorical	2024 2025 2026
Which cohort came earlier? Order matters; distance does not → Ordinal	2024 < 2025 < 2026
Compare observations over time Distance in time matters → Quantitative	John's PhD: 2021–2026 (5 years) Mary's PhD: 2023–2026 (3 years)

Attribute type is determined by semantics and task—not by how values are stored.

Relations come from semantics and can carry attributes

A relation describes how two or more items are related.

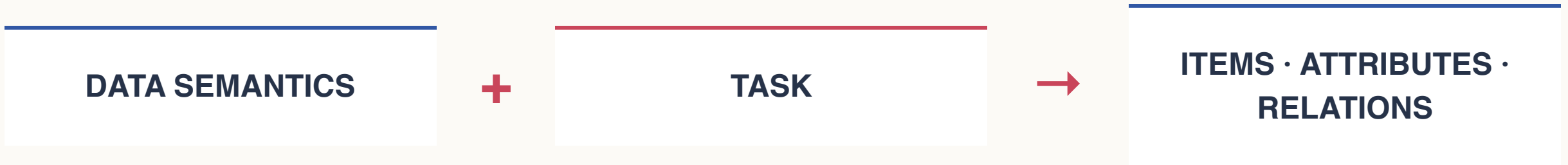
Illustrative Citi Bike trip records

start_station	end_station	duration (min)
1001	1002	12
1005	1001	18
1001	1002	9
1008	1006	22
1002	1005	15
1001	1002	14

How do we obtain a relation?

1. **Identify the items.** For this task, the items are stations.
2. **Obtain the relation.** Data semantics tells us that each row records a bike trip from `start_station` to `end_station`. For station connectivity, group rows by station pair.
3. **Derive relation attributes.** For 1001 to 1002, trip count is 3 and average duration is 11.7 minutes. These properties belong to the relation, not either station.

Semantics and task jointly determine the abstraction



1 What do the values mean?

2 Which entities matter?

3 What are their attributes?

4 What relations exist?

5 What are the attribute types?

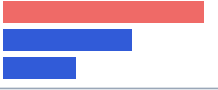
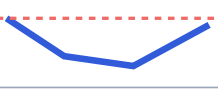
PART 03

Abstract the task

Clarify the comparison before choosing how to show it.

Task abstraction identifies the required comparison

Task abstraction simplifies a domain question into the comparison that a visualization must support.

DOMAIN QUESTION	REQUIRED COMPARISON	CANDIDATE VIEW
Which station has the most departures?	Compare stations and find the maximum	
Did duration return to pre-COVID levels?	Compare a temporal trend with a reference level	
Are duration and distance related?	Compare paired quantitative values	

The required comparison narrows the design space without choosing one final chart.

Different comparisons suggest different chart structures

COMPARISON (action + target)	THE VIEW MUST MAKE IT EASY TO	CANDIDATE CHARTS
Find + extreme value	Compare categories on a common scale	Bar chart, dot plot
Compare + temporal trend	Follow change along an ordered time axis	Line chart
Examine + association	Compare two quantitative values for every item	Scatterplot
Summarize + distribution	Compare frequencies, density, or quantiles	Histogram, box plot
Locate + outlier	See individual values against their context	Dot plot, scatterplot, box plot
Explore + topology	See connections or adjacency among items	Node-link diagram, adjacency matrix

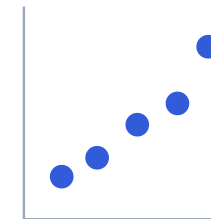
A comparison constrains the chart choice without determining one final chart.

Practice: identify the required comparison

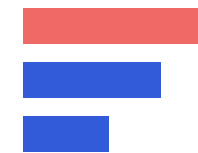
For each question, state the **action + target**, describe the **required comparison**, and match it to a candidate view.

1 • Which station has the most departures?

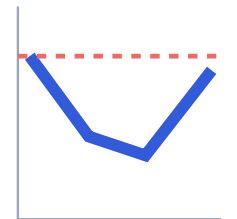
A



B



C



2 • Did trip duration return to pre-COVID levels?

3 • Are duration and distance related?

PART 04

Encode the abstraction visually

Marks represent items and relations; channels carry attributes.

Visual encoding

Visual encoding is the process of converting visual information into a format that the brain can store and process.

In data visualization, marks and channels are fundamental components used to encode and represent information visually.

- **Marks** are basic graphical elements that serve as the building blocks of visual representation. They represent data points or objects in a visualization.
- **Channels** define how data is represented by the appearance of marks. Different channels vary in their effectiveness for encoding different types of data (ordinal and quantitative vs. categorical).
- Proper selection of marks and channels ensures clarity, accuracy, and efficiency in the visual encoding process.

Marks for items

➞ Points



➞ Lines (curves)



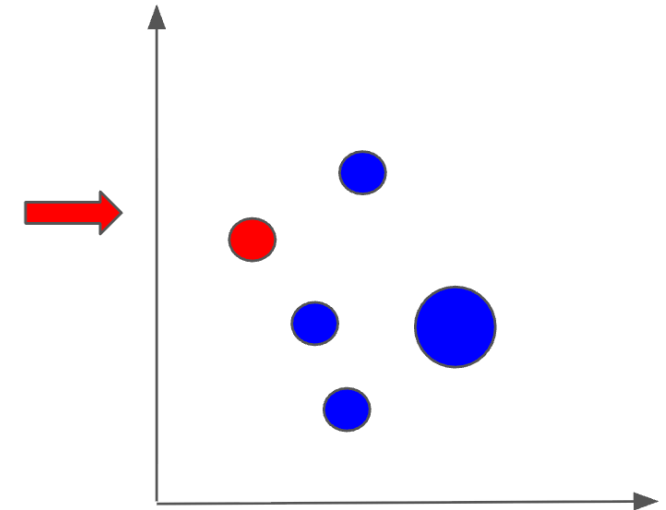
➞ Areas (regions)



Volume is a 3D mark, but it is rarely used.

Example: using point mark

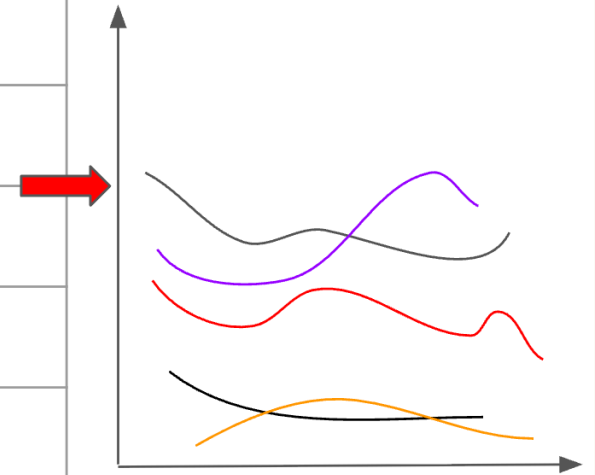
Name	Gender	Weight	height
Kyle	male	30kg	133cm
Stan	male	35kg	135cm
Eric	male	60kg	140cm
Kenny	male	28kg	130cm
Windy	femal	27kg	130cm



Each item is represented by a point.

Example: using line mark (time series)

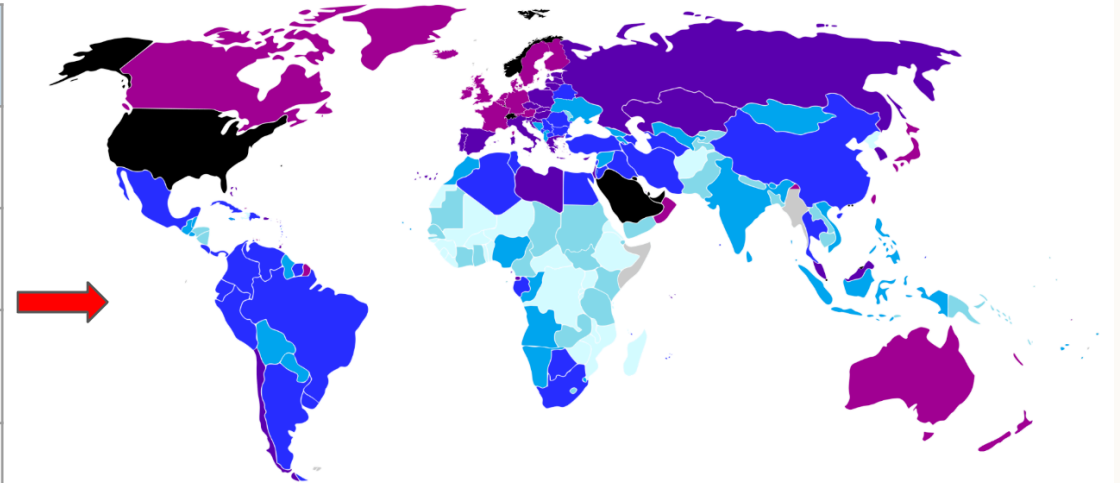
Index	Monday	Tuesday	Wednesday	Thursday	Friday
Dow	10000	100200	9800	10500	10360
Nasdaq	9000	8500	9300	10010	10239
S&P 500	3001	3010	2980	3045	3021
Hang Seng	29010	29300	29601	29105	28800
DAX	13902	14308	13999	14201	14946



Each item is represented by a line.

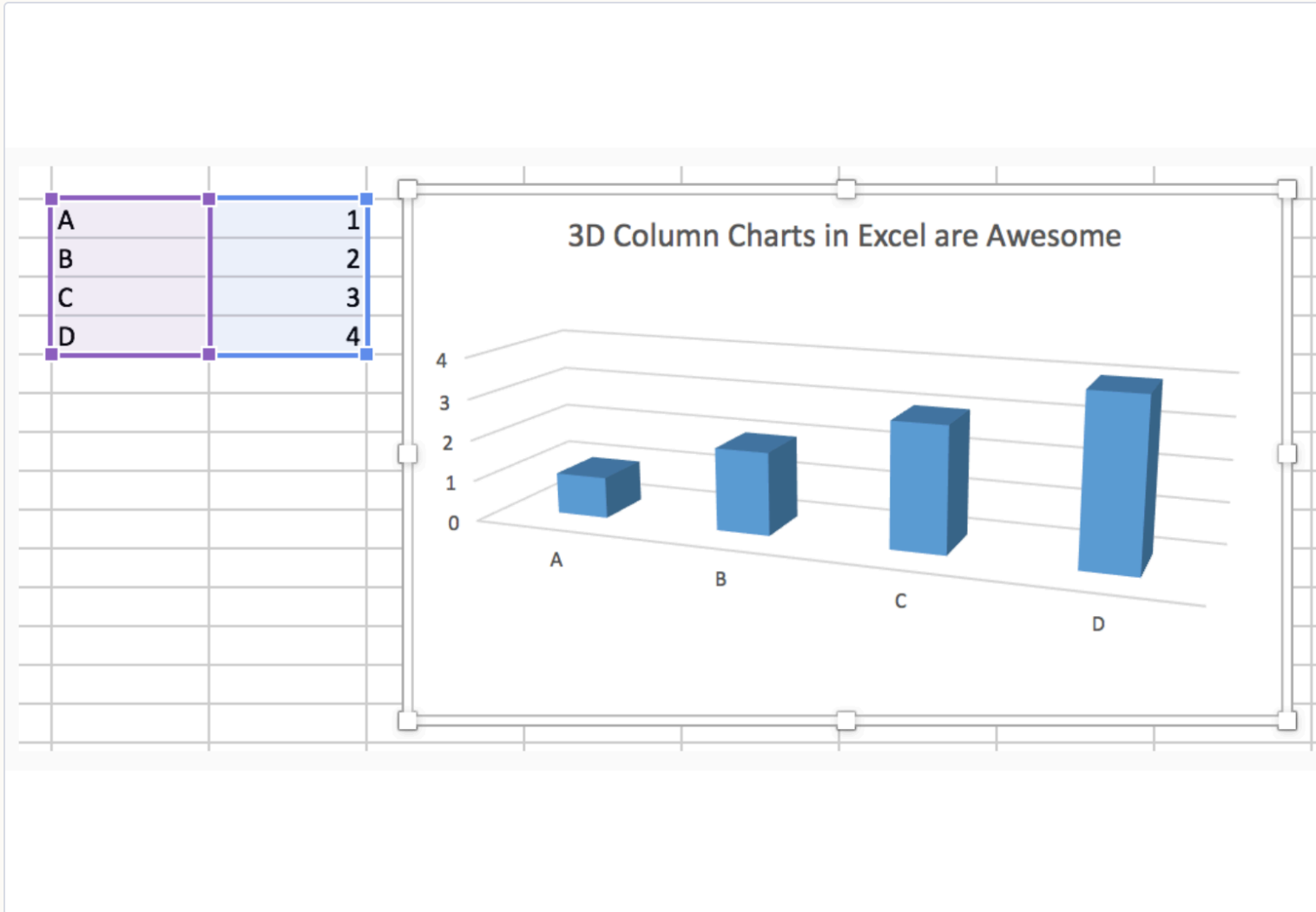
Example: using area mark (map)

Region	GDP	Type
US	10000	Country
EU	9000	Continent
Asia	12000	Continent
China	9000	Country
NYC	1390	City



Each item is represented by an area.

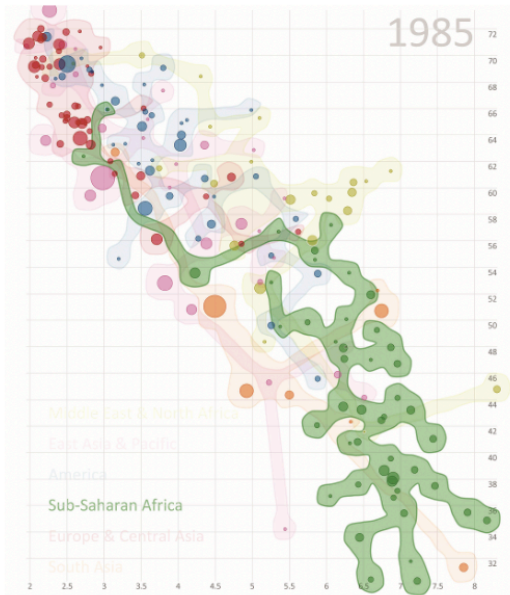
3D mark is not a good choice



A 3D column chart created in Microsoft Excel. The values shown in the bars are 1, 2, 3, and 4. Each column appears to be below to its actual value (refer to the axis guidelines)

Marks for relations (links)

➔ Containment



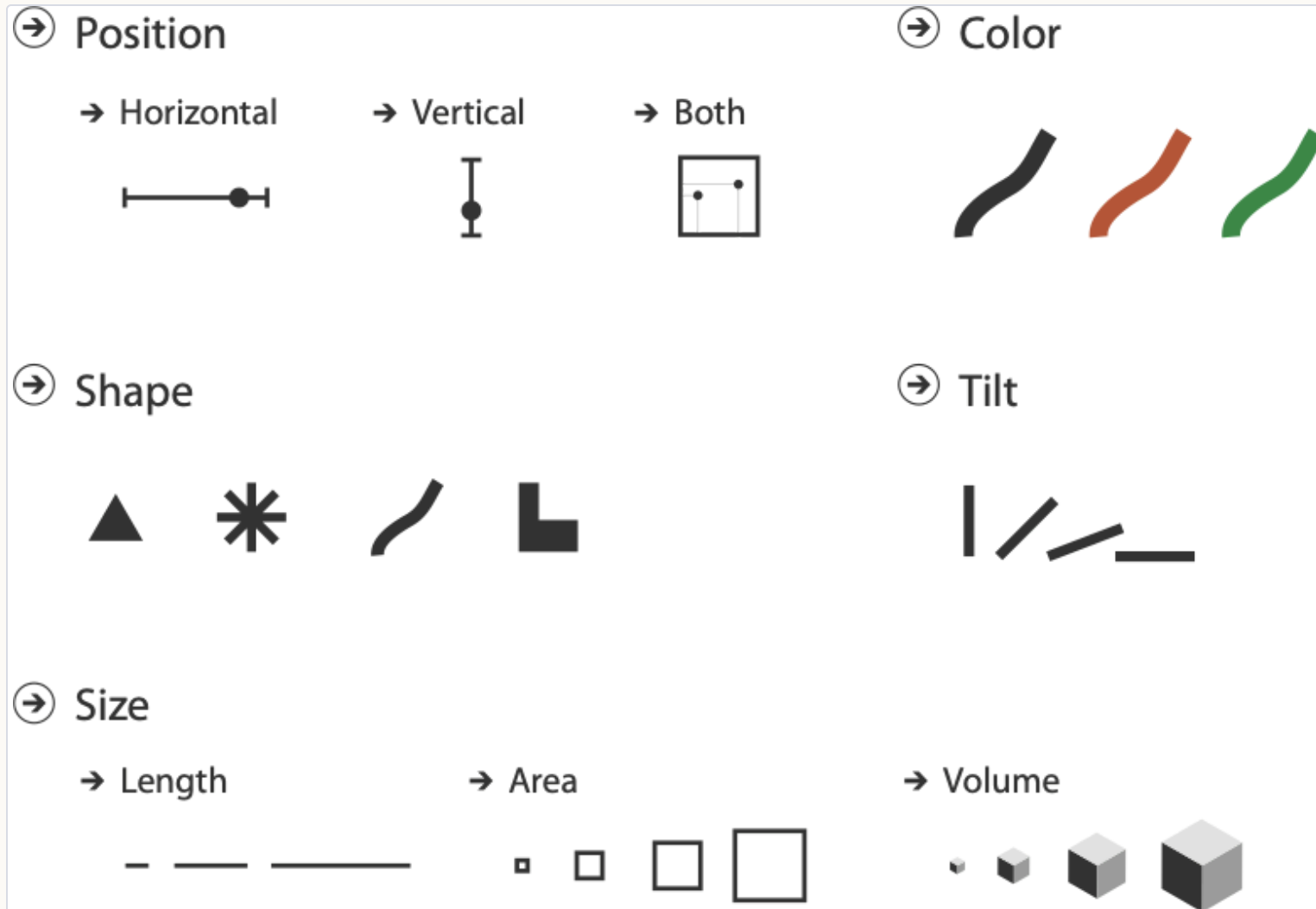
➔ Connection



Items are grouped according to regions or links.

It is an application of Enclosure and Connectedness in Gestalt laws.

Channels

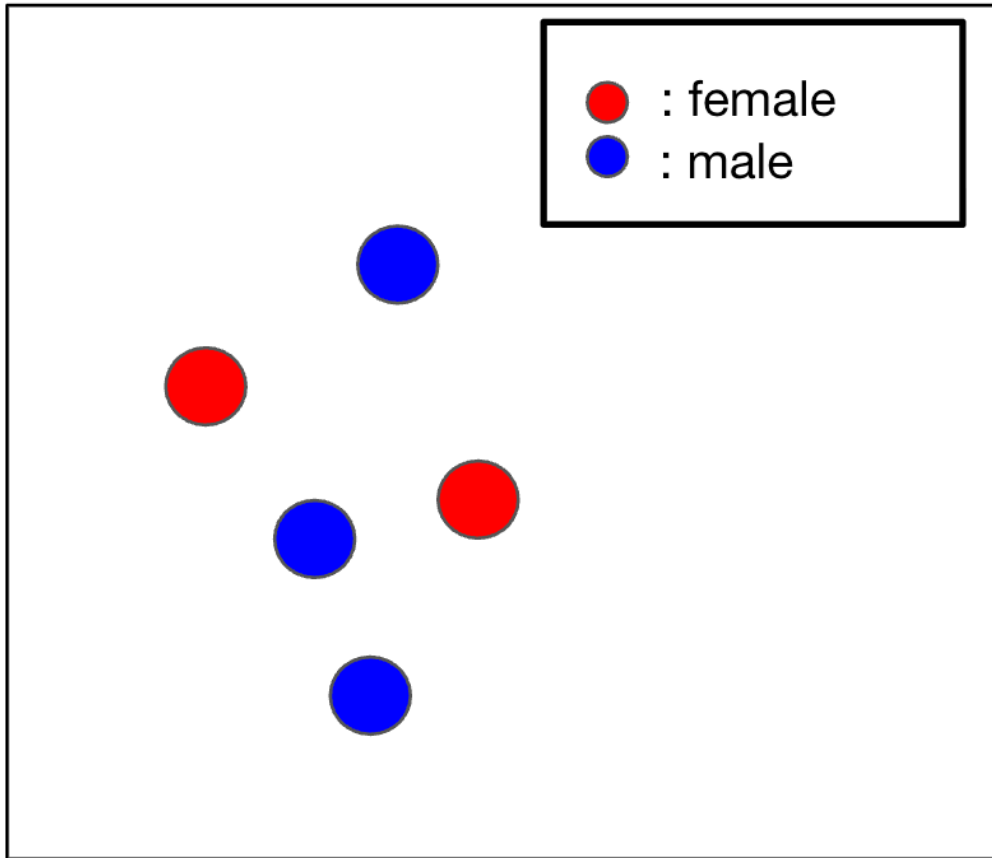


Channels are appearance of marks.

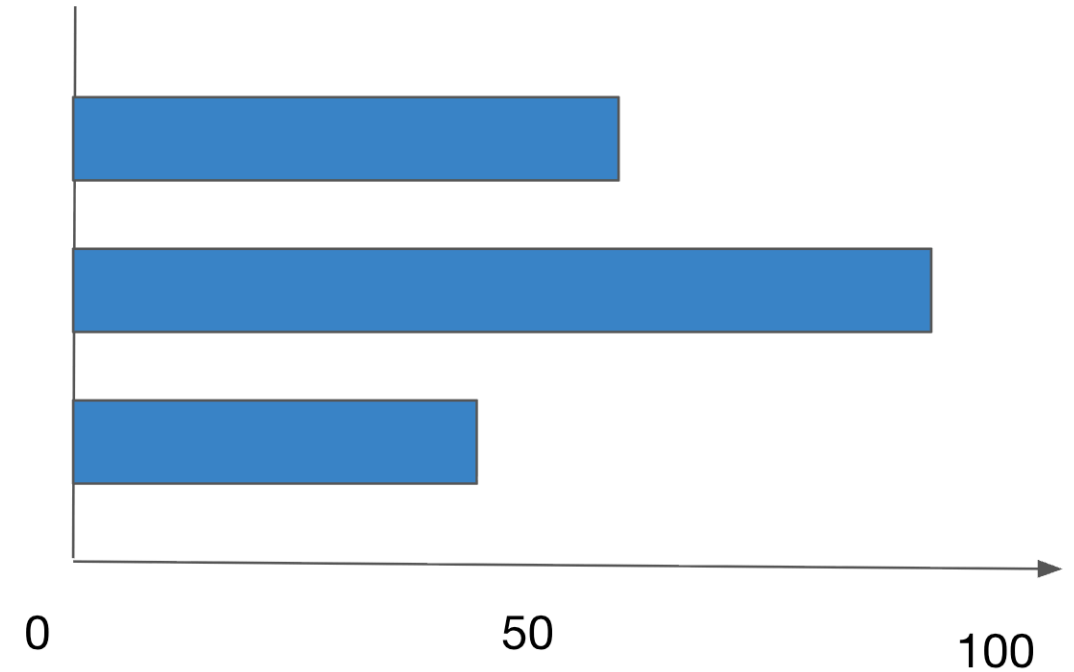
They encode attributes of an item and have two types:

- *Magnitude channels*: show how many
- *Identity channels*: show what

Example: using channels



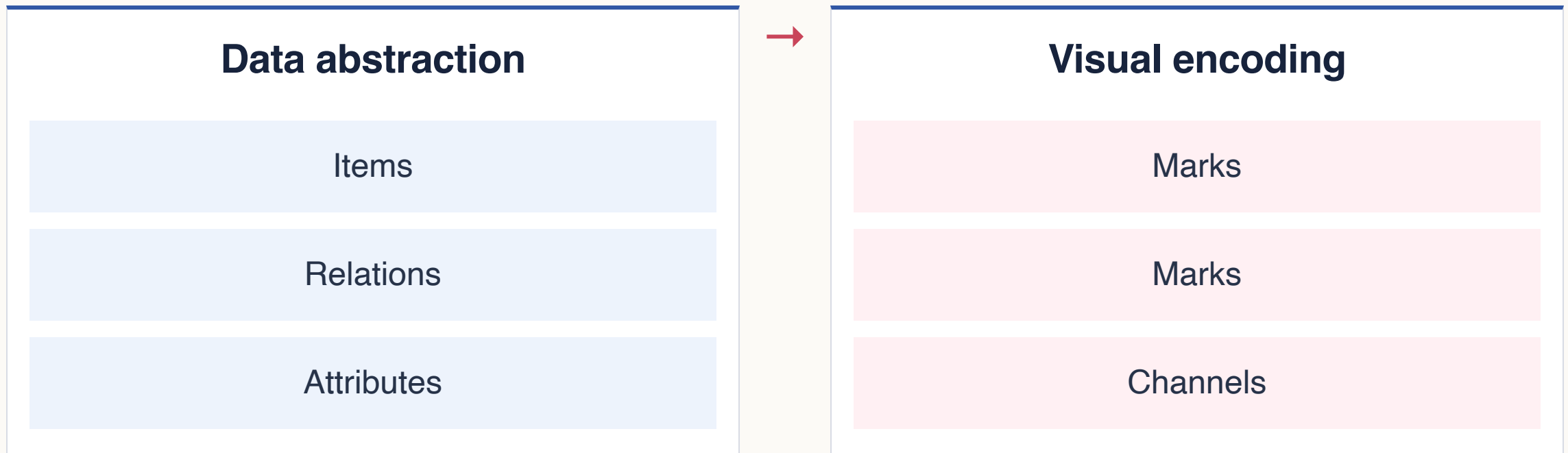
Point marks with the hue and position channels



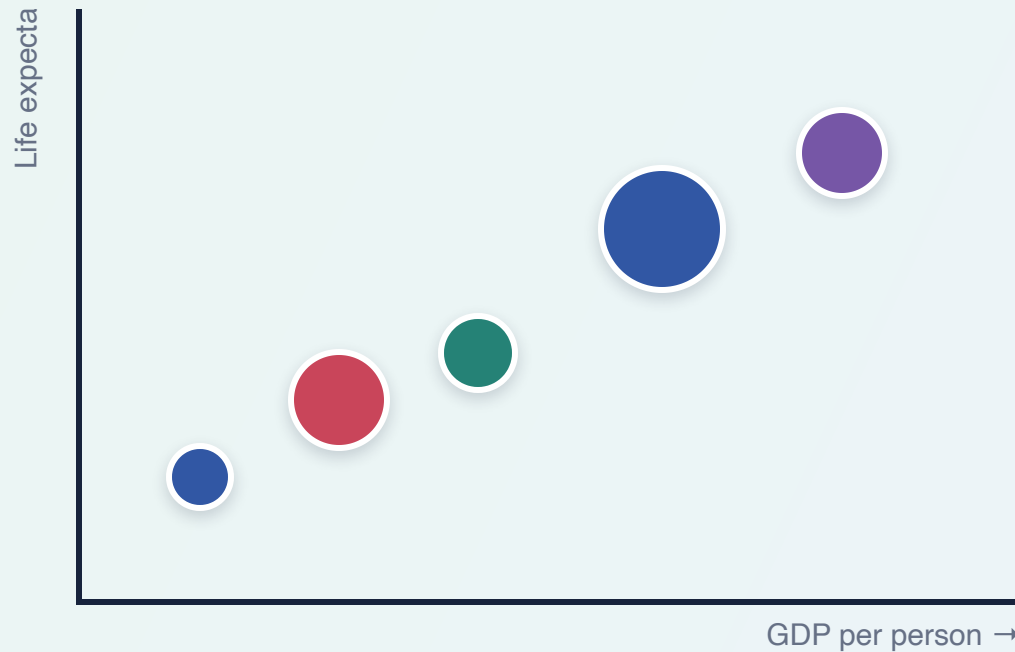
Line marks (bars) with length and spatial region channels

Visual encoding: Map the abstraction to marks and channels

Items / relations → marks · Attributes → channels



The workflow produces a candidate encoding



Raw data → semantics → data abstraction → task
abstraction → visual encoding

Next lecture

Is hue appropriate for continent?

Is size appropriate for population?

Can viewers distinguish the colors?

Do channels interfere?

Remember

Items / relations → marks

Attributes → channels